

Lab Manual — Week 10

Concurrency & Distributed Systems — Lebanese University, Faculty of Engineering

Topic: Network = Concurrency + Failure

Learning Objectives

1. State the three laws of networks.
2. Convert queue semantics into socket protocol messages.
3. Explain why sockets do not share Java objects.
4. Use Toxiproxy to plan latency, bandwidth, reset, and timeout experiments.
5. Summarize throughput, latency, and timeout measurements.

Lab Folder

```
week10-lab/week10-day1-network-concurrency-failure-toxiproxy
```

```
.\gradlew.bat test
.\gradlew.bat studentCheck
.\run-all-demos.ps1
```

Part 1 — Network Laws

Open `THREE_LAWS_OF_NETWORKS.md` and implement `Ex1_ThreeNetworkLaws`.

Part 2 — Socket Queue Protocol

Implement commands for PUT, TAKE, ACK, and EMPTY. Explain that messages cross a socket boundary and objects are not shared.

Part 3 — Toxiproxy Plan

Open `TOXIPROXY_GUIDE.md`. Plan baseline, latency, bandwidth/loss, reset/timeout, and metrics.

Part 4 — Measurement Summary

Complete `MEASUREMENT_SUMMARY.md`. Your summary must include request count, throughput, p95 latency, and timeout count.

Optional Docker

```
docker compose up -d
```

The Java smoke tests do not require Docker.

Bridge to Week 11

Week 11 extends this socket boundary to multi-client servers, serialization, versioning, and peer review.